Impedance:

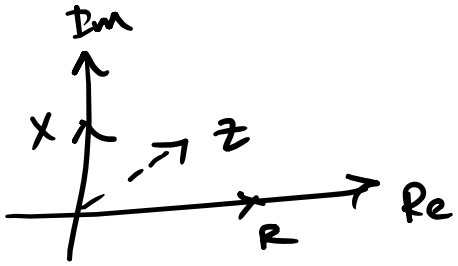
Ratio of phasor voltage to phasor current.

$Z = R + jX$ ← combination of X_L, X_C

$X_L = \omega L$

$X_C = \frac{1}{\omega C}$

Complex impedance plane



$Z = R + j\omega L$

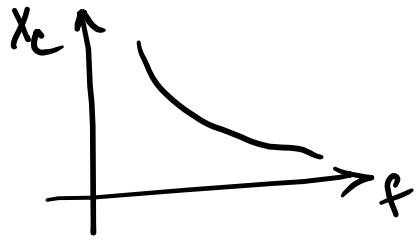
$Z = R + \frac{1}{j\omega C}$

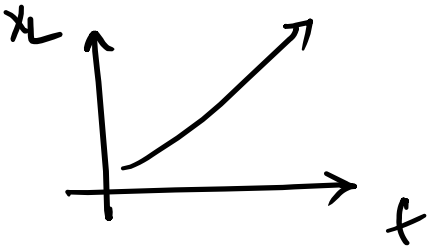
$Z = R + j(X_L - X_C)$

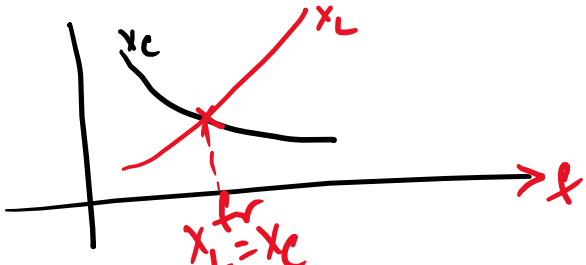
$R = Re(Z)$

$X = Im(Z)$

Resonance:







f_r: Resonant frequency.

Resonance occurs in an AC circuit when both reactances which are equal and opposite cancel each other out.

$X_L = X_C \rightarrow Z = R + j(X_L - X_L)$

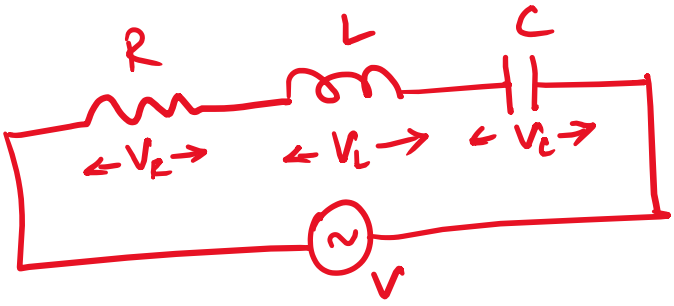
$\boxed{Z = R}$

Complex Impedance plane:

X axis: 'R' → shows energy dissipation.

Yaxis : `X' → shows energy storage.

Series Resonance:



$$Z = R + j\omega L + \frac{1}{j\omega C}$$

$$I = \frac{V}{Z} = \frac{V}{R + j(\omega L - \frac{1}{\omega C})} = \frac{V}{R + j(X_L - X_C)}$$

At Resonance, $Im(Z) = 0$

$$\omega L - \frac{1}{\omega C} = 0 \Rightarrow \omega L = \frac{1}{\omega C}$$

$$\omega_r = \frac{1}{\sqrt{LC}} \quad : f_r = \frac{1}{2\pi\sqrt{LC}}$$

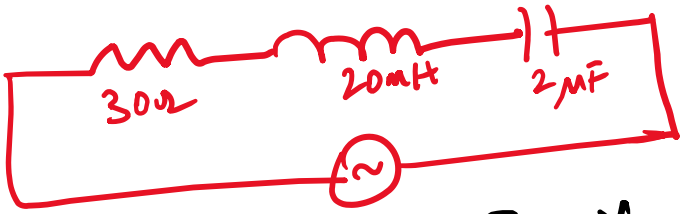
At resonance, $\boxed{I = \frac{V}{R}}$

Ex: Series resonance network, with a resistor 30Ω, a capacitor 2μF and an inductor 20mH connected to a sinusoidal voltage which is constant at 9V for all frequencies.

→ Calc resonant frequency in Hertz.

→ current at resonance.

→ voltage across L, C at resonance.



$X_L = X_C$

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}} = 796\,Hz$$

Current at resonance $I = \frac{V}{R}$

$$= \frac{9}{30} = 300\,mA$$

Voltage across L, C

$$V_L = V_C = I \cdot X_L \text{ or } I \cdot X_C$$

$$= 300mA \times 100 = \underline{\underline{30V}}$$

Series RLC → Calc w_r, f_r

$R = 100\Omega$

$L = 0.1H$

$C = 0.01\mu F$

$w_r = 31622$ rad/s

$f_r = 5032$ Hz.

Two impedances $Z_1 = a + jb$ and $Z_2 = c - jd$ are connected in series. Find the condition for resonance.

$$Z = Z_1 + Z_2 = (a+c) + j(b-d)$$

Condition for resonance $Im(Z) = 0$

$$b - d = 0 \rightarrow \boxed{b = d}$$

Series RLC circuit

$R = 1\Omega$

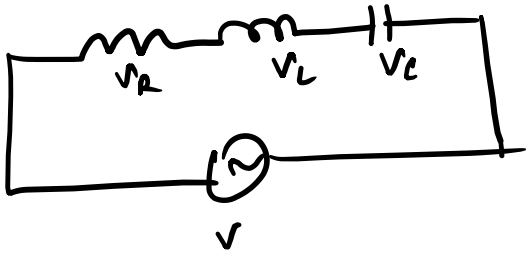
$L = 0.5H$

$C = 0.1\mu F$

→ Calc f_r.

→ 5V sinusoidal at all freq is applied. Draw the phasor diagram showing all voltage and current phasors.

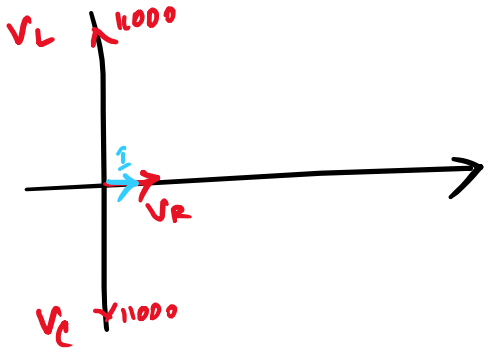
→ Draw phasor for $V = 5V$ @ 750Hz



$$f_r = \frac{1}{2\pi\sqrt{LC}} = 711.76 \text{ Hz}.$$

$$I = \frac{V}{R} = 5A. \; ; V_R = 5V$$

$$V_L = V_C = I \cdot X_L = 11180V$$



$V = 5V$, $f = 750Hz$.

$X_L = \omega L = 2356.2$ $\quad Z = R + j(X_L - X_C)$

$X_C = \frac{1}{\omega C} = 2122.06$ $\quad Z = 1 + j(234.13)$

$$I = \frac{V}{Z} = \frac{5}{1 + j(234.13)}$$

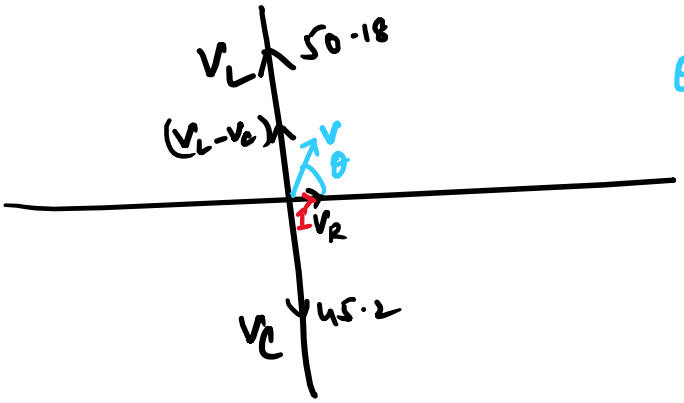
$$\text{mag}(I) = \frac{5}{\sqrt{1^2 + 234.13^2}} = 0.021\,A$$

$$\text{phase}(I) = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = 89.75^\circ$$

$V_R = 0.02V$

$V_L = I \cdot \omega L = 50.18V$

$V_C = I \cdot \frac{1}{\omega C} = 45.2V$



$\theta = 89.75^\circ$